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Moment swaps

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In this paper we discuss moment swaps. These derivatives depend on the realized higher moments of the underlying. A special case is the nowadays popular variance swaps. After introducing moment swaps we discuss how to hedge these derivatives. Moreover, we show how the classical hedge of the variance swap in terms of a position in log-contracts and a dynamic trading strategy can be significantly enhanced by using third moment swaps.

Keywords: Moment derivatives; Stochastic volatility models; Hedging

1. Introduction

This paper deals with moment swaps. These derivatives depend on the realized higher moments of the underlying. More precisely, their payoff is a function of powers of the (daily) log-returns and give you protection against different kinds of market *shocks*. In the case of squared log-returns we have the so-called variance swaps. Variance swaps are nowadays liquidly traded for several underlyers and were created to cover against changes in the volatility regime. Besides the latter, skewness, kurtosis, etc. also play an important role. To protect against a wrongly estimated skewness or kurtosis, moment derivatives of higher order can be useful. Moreover, we will show that the classical hedge of the variance swap using log-contracts can be significantly improved using moment swaps.

Furthermore, recent studies (Nualart and Schoutens 2000, 2001, Corcuera *et al.* 2004, 2005) suggest that functionals of powers of returns seem the natural choice to complete the market. It was shown for example in Corcuera *et al.* (2004, 2005) that under an incomplete Lévy market allowing trade in the power assets of all orders leads to a complete market. Power assets are strongly related to the realized higher moments and in a discrete time framework, they mainly coincide (see Corcuera *et al.* (2005)).

This paper is organized as follows. In the next section we introduce moment swaps on stocks and futures. After linking the prices between the two cases, we discuss the hedging of higher order moment swaps. Basically, the hedge consists of a position in the so-called log-contract, dealt with in more detail in section 4, a dynamic trading strategy and a static position in moment swaps of lower order. In section 3, we introduce some stochastic volatility models (Heston and time-changed Lévy models). We give closed-formula expressions for the price of the log-contract and assess the hedging of the nowadays popular variance swap using a 3rd-order moment swap. We will show that the classical hedge of a variance swap in terms of a static position in the log-contract and a dynamic trading strategy can be significantly enhanced by taking a position in third moment swaps by running a simulation study under the above-mentioned stochastic volatility models.

2. Moment swaps

2.1. Introducing moment swaps

Consider a stock (index), which is assumed to pay out a continuous dividend-yield $q \geq 0$. Denote by $S = \{S_t, t \geq 0\}$ the price process of the stock. Assume we have also available the classical bank-account/bond, with a compound interest rate r. The bond price process is hence given by $B = \{B_t = \exp(rt), t \geq 0\}$.

Consider a finite set of equally spaced discrete times $\{t_i = i\Delta t, i = 0, 1, \ldots, n\}$, with $\Delta t = T/n$ at which the path of the underlying is monitored. We denote the price of the underlying at these points, i.e. S_{t_i}, by S_i

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for simplicity. Typically, $\{t_0 = 0, t_1 = \Delta t, t_2 = 2\Delta t, \ldots, t_n = T\}$ corresponds to daily closing times and S_i is the closing price at day i. Note that then

$$\log(S_i) - \log(S_{i-1}), \quad i = 1, \ldots, n,$$

correspond to the daily log-returns.

We assume that futures are available on S which expire at time T. By risk-neutral valuation, the price process of the futures is given by $F = \{F_t = \exp((r-q)\times(T-t))S_t\}$. In line with the above notation, let us write $F_i = F_{t_i}$.

Next, we define the *moment swaps* on the stock. The kth-moment swap is a contract where the parties agree to exchange the amount

$$MOMS^{(k)}_{\text{stock}} = N \times \left(\sum_{i=1}^{n} (\log(S_i) - \log(S_{i-1}))^k\right) = N \times \left(\sum_{i=1}^{n} \left(\log\left(\frac{S_i}{S_{i-1}}\right)\right)^k\right),$$

at maturity T, where N is the nominal amount, which we take for simplicity from now on equal to $N = 1$ for all swaps considered.

A special case of these swaps is the 2nd-moment swap, better known as the variance swap (VS). The payoff function in that case is given by

$$VS = \sum_{i=1}^{n} (\log(S_i) - \log(S_{i-1}))^2.$$

Basically this contract swaps fixed (annualized) variance (second moment) with the realized variance (second moment) and as such provides protection against unexpected or unfavourable changes in volatility. Higher moment swaps provide the same kind of protection. The $MOMS^{(3)}$ is related to realized skewness and provides protection against changes in the symmetry of the underlying distribution. $MOMS^{(4)}$ derivatives are linked to realized kurtosis and provide protection against the unexpected occurrences of very large jumps, or in other words changes in the tail behaviour of the underlying distribution.

It will turn out to be quite convenient to first consider moment swaps with the futures price as the underlying. One can make easily the transformation to the case where the stock price is taken as the underlying for the moment derivative. The kth-moment swap on the futures is a contract where the parties agree to exchange the amount

$$MOMS^{(k)}_{\text{futures}} = \sum_{i=1}^{n} (\log(F_i) - \log(F_{i-1}))^k = \sum_{i=1}^{n} \left(\log\left(\frac{F_i}{F_{i-1}}\right)\right)^k,$$

at maturity T.

The relation between the futures and the stock moment swaps can be easily deduced using the identity $F_i = \exp((r-q)(T-t_i))S_i$. Indeed,

$$\begin{aligned} MOMS^{(k)}_{\text{futures}} &= \sum_{i=1}^{n} \left(\log\left(\frac{F_i}{F_{i-1}}\right)\right)^k \\ &= \sum_{i=1}^{n} \left(-(r-q)\Delta t + \log\left(\frac{S_i}{S_{i-1}}\right)\right)^k \\ &= \sum_{i=1}^{n} \sum_{j=0}^{k} \binom{k}{j} (-(r-q)\Delta t)^j \left(\log\left(\frac{S_i}{S_{i-1}}\right)\right)^{k-j} \\ &= \sum_{j=0}^{k-2} \binom{k}{j} (-(r-q)\Delta t)^j MOMS^{(k-j)}_{\text{stock}} - (r-q)T \\ &\quad \times (-(r-q)\Delta t)^{k-1} + k(-(r-q)\Delta t)^{k-1} LOG_{\text{stock}}, \end{aligned}$$

where due to telescoping

$$LOG_{\text{stock}} = \sum_{i=1}^{n} \log\left(\frac{S_i}{S_{i-1}}\right) = \log(S_T) - \log(S_0)$$

is the so-called *log-contract* (on the stock). This log-contract will also play an important role later on in the hedging of moment-swaps. Ignoring higher order powers of Δt, which is assumed to be small, the above calculation leads to the following approximation:

$$MOMS^{(k)}_{\text{futures}} \approx MOMS^{(k)}_{\text{stock}} - k((r-q)\Delta t) MOMS^{(k-1)}_{\text{stock}}. \tag{1}$$

2.2. Hedging moment swaps

We focus on hedging the moment swaps which are written on the futures price as underlying.

In line with the results obtained by Carr and Lewis (2004), first consider the following (Taylor-like) expansion of the kth power of the logarithmic function

$$(\log(x))^k = k!\left(x - 1 - \log(x) - \frac{(\log(x))^2}{2!} - \frac{(\log(x))^3}{3!} - \cdots - \frac{(\log(x))^{k-1}}{(k-1)!} + \mathcal{O}((x-1)^{k+1})\right). \tag{2}$$

This formula can be easily obtained from the power series representation of the exponential function. Indeed, we have

$$\exp(y) = 1 + y + \frac{y^2}{2!} + \cdots + \frac{y^k}{k!} + \mathcal{O}(y^{k+1}).$$

Substituting y by $\log(x)$, rearranging terms and noting that $\mathcal{O}(\log(x)^{k+1}) = \mathcal{O}((x-1)^{k+1})$ leads to (2).

Next, if we substitute in (2) x by F_i/F_{i-1}, we obtain

$$(\log(F_i/F_{i-1}))^k = k!\left(\frac{\Delta F_i}{F_{i-1}} - \log(F_i/F_{i-1}) - \sum_{j=2}^{k-1} \frac{(\log(F_i/F_{i-1}))^j}{j!} + \mathcal{O}((\Delta F_i/F_{i-1})^{k+1})\right),$$

where $\Delta F_i = F_i - F_{i-1}$. Summing over i gives a decomposition of the $MOMS^{(k)}_{\text{futures}}$ payoff:

$$\begin{aligned} MOMS^{(k)}_{\text{futures}} &= \sum_{i=1}^{n} (\log(F_i/F_{i-1}))^k \\ &= k! \sum_{i=1}^{n} \left(\frac{\Delta F_i}{F_{i-1}} - \log(F_i/F_{i-1}) - \sum_{j=2}^{k-1} \frac{(\log(F_i/F_{i-1}))^j}{j!} \right. \\ &\quad \left. + \mathcal{O}((\Delta F_i/F_{i-1})^{k+1}) \right) \\ &= -k!(\log(F_T) - \log(F_0)) + k! \sum_{i=1}^{n} \frac{\Delta F_i}{F_{i-1}} \\ &\quad - \sum_{j=2}^{k-1} \frac{k!}{j!} MOMS^{(j)}_{\text{futures}} + \mathcal{O}\left(\sum_{i=1}^{n} (\Delta F_i/F_{i-1})^{k+1} \right) \end{aligned} \tag{3, 4}$$

due to telescoping. Thus up to $(k+1)$th-order terms the sum of the kth powered log-returns decomposes into

(a) the payout from $-k!$ log-contracts on the futures with payoff

$$LOG_{\text{futures}} = \log(F_T) - \log(F_0) = LOG_{\text{stock}} - (r-q)T;$$

(b) a dynamic strategy ($k! \sum_{i=1}^{n} (\Delta F_i/F_{i-1})$) in futures;
(c) a series of moment contracts of order strictly smaller than k.

The log-contract can be hedge with a purely static position in bonds and vanilla options (see Carr and Lewis (2004)).

For $k=2$, equation (4) leads to the classical hedge of the variance swap in terms of log-contracts:

$$VS_{\text{futures}} = MOMS^{(2)}_{\text{futures}} \approx 2\left(\sum_{i=1}^{n} \frac{\Delta F_i}{F_{i-1}} - LOG_{\text{futures}} \right). \tag{5}$$

Hence hedging the variance swap comes down into a short position of 2 logcontracts and following a dynamic strategy (which can be entered at zero costs) in terms of futures. The performance of this hedge will be discussed in the next section.

For $k=3$, equation (4) leads to

$$MOMS^{(3)}_{\text{futures}} \approx 6\left(\sum_{i=1}^{n} \frac{\Delta F_i}{F_{i-1}} - LOG_{\text{futures}} - \frac{1}{2} VS_{\text{futures}} \right).$$

Rearranging terms results in

$$VS_{\text{futures}} \approx 2\left(\sum_{i=1}^{n} \frac{\Delta F_i}{F_{i-1}} - LOG_{\text{futures}} \right) - \frac{1}{3} MOMS^{(3)}_{\text{futures}}. \tag{6}$$

Additionally shorting 1/3 third moment swaps, $MOMS^{(3)}_{\text{futures}}$, will thus lead to an improvement of the hedging strategy of the variance swap. In section 5 we quantify this improvement.

3. Some advanced models

In this section, we consider some advanced models under which we will try to assess the hedging performance of the variance swap using moment derivatives. We opt to focus especially on the Heston model and Lévy models with stochastic time because they give very satisfactory global calibration fits and are nowadays finding their way into the derivatives world. We completely follow the notation of Schoutens *et al.* (2004) (see also Schoutens *et al.* (2005)). For more details we refer to Schoutens (2003). Here, we will give the dynamics of the model in terms of the characteristic function $\phi(u,t)$ of the logarithm of the price process $\log S_t$, i.e.

$$\phi(u,t) = E[\exp(\mathrm{i}u \log(S_t))].$$

3.1. The Heston stochastic volatility model

The stock price process in the Heston stochastic volatility model (HEST) follows the Black–Scholes (1973) SDE in which the volatility is behaving stochastically over time:

$$\frac{\mathrm{d}S_t}{S_t} = (r-q)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad S_0 \geq 0,$$

with the (squared) volatility following the classical Cox–Ingersoll–Ross (CIR) process:

$$\mathrm{d}\sigma_t^2 = \kappa(\eta - \sigma_t^2)\mathrm{d}t + \theta\sigma_t \mathrm{d}\tilde{W}_t, \quad \sigma_0 \geq 0,$$

where $W = \{W_t, t \geq 0\}$ and $\tilde{W} = \{\tilde{W}_t, t \geq 0\}$ are two correlated standard Brownian motions such that $\mathrm{Cov}[\mathrm{d}W_t \mathrm{d}\tilde{W}_t] = \rho \mathrm{d}t$.

The characteristic function $\phi(u,t)$ is in this case given by (see Heston (1993) or Bakshi *et al.* (1997)):

$$\begin{aligned} \phi(u,t) &= E[\exp(\mathrm{i}u \log(S_t)) | S_0, \sigma_0^2] \\ &= \exp(\mathrm{i}u(\log S_0 + (r-q)t)) \\ &\quad \times \exp(\eta\kappa\theta^{-2}((\kappa - \rho\theta u\mathrm{i} - d)t \\ &\quad - 2\log((1 - g\mathrm{e}^{-dt})/(1-g)))) \\ &\quad \times \exp(\sigma_0^2\theta^{-2}(\kappa - \rho\theta\mathrm{i}u - d)(1 - \mathrm{e}^{-dt})/(1 - g\mathrm{e}^{-dt})), \end{aligned}$$

where

$$d = ((\rho\theta u\mathrm{i} - \kappa)^2 - \theta^2(-\mathrm{i}u - u^2))^{1/2}, \tag{7}$$

$$g = (\kappa - \rho\theta u\mathrm{i} - d)/(\kappa - \rho\theta u\mathrm{i} + d). \tag{8}$$

3.2. Lévy models with stochastic time

It is generally accepted that one of the more practical improvements one may make in the light of the well-known shortfalls of the classical Black–Scholes model is to replace the underlying source of randomness, a Brownian motion, by a Lévy process (VG, NIG, ...). Here we go one step further and will work with models which also incorporate a stochastic volatility effect. The fitting/calibration abilities of these Lévy models

with stochastic time/volatility are very satisfactory (Schoutens 2003).

The Lévy models with stochastic time considered in this paper are built out of two independent stochastic processes as in Carr *et al.* (2003). This technique dates back to Mandelbrot and Taylor (1967). The first process is a Lévy process. The behaviour of the asset price will be modelled by the exponential of the Lévy process, $X = \{X_t, t \geq 0\}$ suitably time-changed, by a second process, a stochastic clock, that builds in a stochastic volatility effect by making time stochastic.

The economic time elapsed in t units of calendar time will be given by the integrated process $Y = \{Y_t, t \geq 0\}$

$$Y_t = \int_0^t y_s \mathrm{d}s$$

of a subordinator (a positive process), $y = \{y_t, t \geq 0\}$. Let us denote by $\varphi(u; t, y_0)$ the characteristic function of Y_t given y_0. The (risk-neutral) price process $S = \{S_t, t \geq 0\}$ is now modelled as follows:

$$S_t = S_0 \frac{\exp((r-q)t)}{E[\exp(X_{Y_t})|y_0]} \exp(X_{Y_t}), \tag{9}$$

where $X = \{X_t, t \geq 0\}$ is a Lévy process. The factor $\exp((r-q)t)/E[\exp(X_{Y_t})|y_0]$ puts us immediately into the risk-neutral world by a mean-correcting argument. Basically, we model the stock price process as the ordinary exponential of a time-changed Lévy process. The process incorporates jumps (through the Lévy process X_t) and stochastic volatility (through the time change Y_t). The characteristic function $\phi(u, t)$ for the log of our stock price is given by

$$\begin{aligned}\phi(u, t) &= E[\exp(\mathrm{i}u \log(S_t))|S_0, y_0] \\ &= \exp(\mathrm{i}u((r-q)t + \log S_0)) \frac{\varphi(-\mathrm{i}\psi_X(u); t, y_0)}{\varphi(-\mathrm{i}\psi_X(-\mathrm{i}); t, y_0)^{\mathrm{i}u}},\end{aligned} \tag{10}$$

where

$$\psi_X(u) = \log E[\exp(\mathrm{i}uX_1)]\,; \tag{11}$$

$\psi_X(u)$ is called the characteristic exponent of the Lévy process.

Here we consider the CIR process for y and two possible Lévy processes: the NIG and VG process.

3.2.1. NIG Lévy process. A NIG process is based on the normal inverse Gaussian (NIG) distribution, NIG(α, β, δ), with parameters $\alpha > 0$, $-\alpha < \beta < \alpha$ and $\delta > 0$. Its characteristic function is given by

$$\phi_{\mathrm{NIG}}(u; \alpha, \beta, \delta) = \exp\left(-\delta\left(\sqrt{\alpha^2 - (\beta + \mathrm{i}u)^2} - \sqrt{\alpha^2 - \beta^2}\right)\right).$$

Since this is an infinitely divisible characteristic function, one can define the NIG-process $X^{(\mathrm{NIG})} = \{X_t^{(\mathrm{NIG})}, t \geq 0\}$, with $X_0^{(\mathrm{NIG})} = 0$, as the process having stationary and independent NIG distributed increments. An increment over the time interval $[s, s+t]$ follows a NIG$(\alpha, \beta, \delta t)$ law.

3.2.2. VG Lévy process. The characteristic function of the VG(C, G, M) distribution (see e.g. Madan *et al.* (1998)), with parameters $C > 0$, $G > 0$ and $M > 0$ is given by

$$\phi_{\mathrm{VG}}(u; C, G, M) = \left(\frac{GM}{GM + (M-G)\mathrm{i}u + u^2}\right)^C.$$

This distribution is infinitely divisible and one can define the VG-process $X^{(\mathrm{VG})} = \{X_t^{(\mathrm{VG})}, t \geq 0\}$ as the process which starts at zero, has independent and stationary increments and where the increment $X_{s+t}^{(\mathrm{VG})} - X_s^{(\mathrm{VG})}$ over the time interval $[s, s+t]$ follows a VG(Ct, G, M) law.

3.2.3. CIR stochastic clock. Carr *et al.* (2003) use as the rate of time change the CIR process that solves the SDE:

$$\mathrm{d}y_t = \kappa(\eta - y_t)\mathrm{d}t + \lambda y_t^{1/2}\mathrm{d}W_t,$$

where $W = \{W_t, t \geq 0\}$ is a standard Brownian motion. The characteristic function of Y_t (given y_0) is explicitly known (see Cox *et al.* (1985)):

$$\begin{aligned}&\varphi_{CIR}(u, t; \kappa, \eta, \lambda, y_0) \\ &\quad = E[\exp(\mathrm{i}uY_t)|y_0] \\ &\quad = \frac{\exp(\kappa^2\eta t/\lambda^2)\exp(2y_0\mathrm{i}u/(\kappa + \gamma\coth(\gamma t/2)))}{(\cosh(\gamma t/2) + \kappa\sinh(\gamma t/2)/\gamma)^{2\kappa\eta/\lambda^2}},\end{aligned}$$

where

$$\gamma = \sqrt{\kappa^2 - 2\lambda^2\mathrm{i}u}.$$

4. The log-contract

In section 2, the log-contract played a very important role. We note here that, under a lot of models, closed expressions for the price of the log-contract is available. Indeed, e.g. we have under the Black–Scholes model with σ as usual denoting the volatility parameter,

$$\begin{aligned}LOG_{\mathrm{stock}} &= \exp(-rT)E_Q[\log(S_T) - \log(S_0)|\mathcal{F}_0] \\ &= \exp(-rT)E_Q[(r - q - \sigma^2/2)T + \sigma W_T|\mathcal{F}_0] \\ &= \exp(-rT)(r - q - \sigma^2/2)T.\end{aligned}$$

For the more advanced models, the price of the log-contract can be easily obtained from the (risk-neutral) characteristic function of $\log(S_T)$, i.e.

$$\phi(u, t) = E_Q[\exp(\mathrm{i}u\log(S_T)|\mathcal{F}_0].$$

Then,

$$-\mathrm{i}\frac{\partial\phi(u, t)}{\partial u}\bigg|_{u=0} = E_Q[\log(S_T)|\mathcal{F}_0].$$

Hence

$$\begin{aligned}LOG_{\mathrm{stock}} &= \exp(-rT)E_Q[\log(S_T) - \log(S_0)|\mathcal{F}_0] \\ &= \exp(-rT)\left(-\mathrm{i}\frac{\partial\phi(0, t)}{\partial u} - \log(S_0)\right).\end{aligned}$$

Doing this exercise for the Heston model, we obtain

$$LOG_{\text{stock}}^{(HEST)} = \frac{\exp(-rT)}{2\kappa}\big(2\kappa(r-q)T - \eta\kappa T - \eta e^{-\kappa T} + \eta - \sigma_0^2 + \sigma_0^2 e^{-\kappa T}\big).$$

For the case of the time-changed Lévy models, with $\phi(u,t)$ as in (10), this comes down to

$$\begin{aligned} LOG_{\text{stock}} &= \exp(-rT)\big((r-q)T - \varphi'(0)\psi_X'(0) - \log(\varphi(-i\psi_X(i)))\big) \\ &= \exp(-rT)\big((r-q)T + E[Y_T]E[X_1] - \log(E[\exp(X_{Y_T})])\big), \end{aligned}$$

where ψ_X is as in (11).

Table 1. LOG_{stock} prices.

Model	$LOG_{\text{stock}} \times 10000$
HEST	−32.64
VG-CIR	−48.33
NIG-CIR	−48.30

Table 2. Parameters.

HEST
$\sigma_0^2 = 0.0654, \kappa = 0.6067, \eta = 0.0707, \theta = 0.2928, \rho = -0.7571$
VG-CIR
$C = 18.0968, G = 20.0276, M = 26.3971, \kappa = 1.2145,$
$\eta = 0.5501,\ \lambda = 1.7913, y_0 = 1$
NIG-CIR
$\alpha = 16.1975, \beta = -3.1804, \delta = 1.0867, \kappa = 1.2101, \eta = 0.5507,$
$\lambda = 1.7864, y_0 = 1$

We obtain the prices in table 1, by applying the above formulas for the models with parameters given in table 2. Note that the parameters in table 2 were obtained, as described in Schoutens *et al.* (2004), from a global calibration procedure on the same volatility surface (of the EuroStoxx50). All models fitted this surface quite well and hence vanilla prices under the different models were almost equal. However, as already reported in Schoutens *et al.* (2004) (and Schoutens *et al.* (2005)), exotic prices can differ quite significantly and one has a clear issue of model risk. This is again illustrated here, the log-contract prices differ by almost 50% in some cases.

5. Variance swap hedging using moment swaps

In this section we compare the performance of the hedging of a variance swap. We compare the strategy without moment swaps described in (5) with the strategy including a 3rd moment swap as in (6) and this for the different models described in section 3. The parameters are again taken from Schoutens *et al.* (2004) and given in table 2.

In figures 1–3 one sees the Profit and Loss (P&L) distribution for the hedging of 10 000 variance swap contracts. Each P&L distribution was obtained by running a million scenarios. Table 3 overviews the mean and the standard deviation of the P&L distribution. μ_1 and σ_1 corresponds to respectively the mean and standard deviation of the hedging without moment derivatives as in (5); μ_2 and σ_2 refers to the case including a 3rd-order moment swap as in (6).

As expected from a theoretical point of view, here we also empirically observe that the introduction of a 3rd

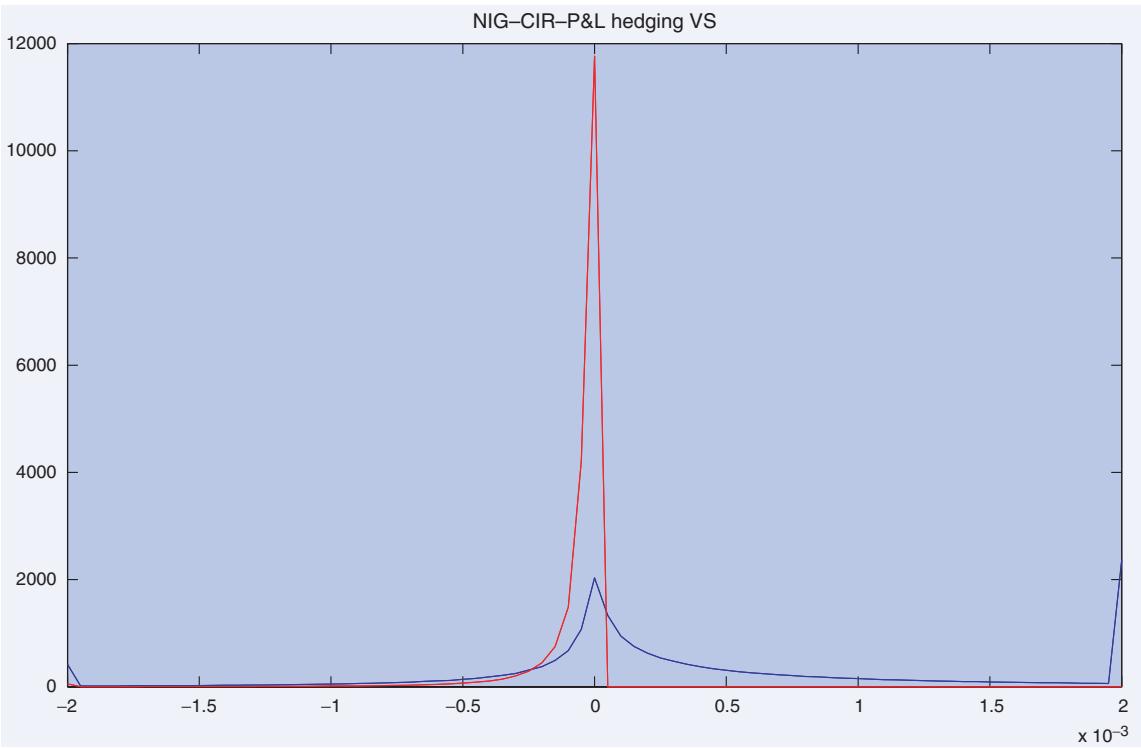

Figure 1. Variance swap hedge P&L under NIG-CIR. Hedge (5) in blue; hedge (6) in red.

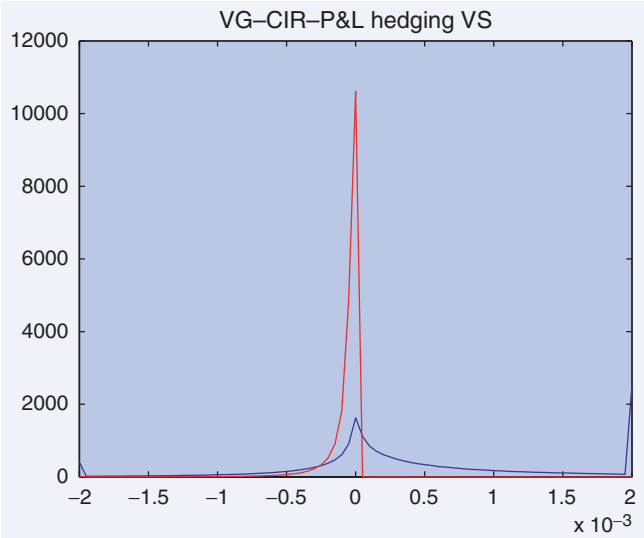

Figure 2. Variance swap hedge P&L under VG-CIR. Hedge (5) in blue; hedge (6) in red.

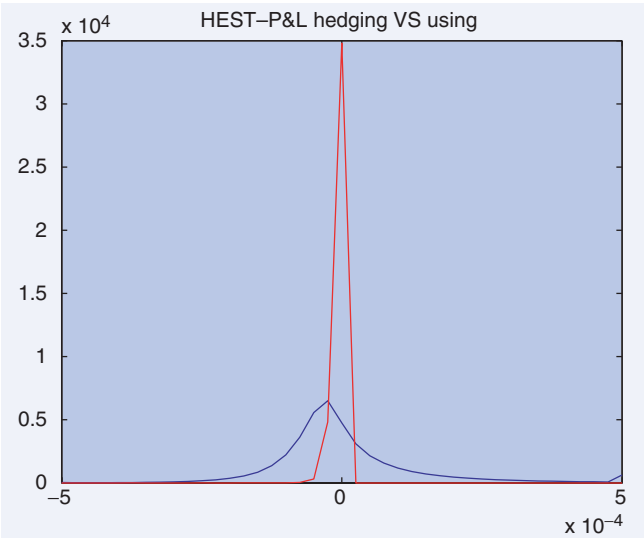

Figure 3. Variance swap hedge P&L under HEST. Hedge (5) in blue; hedge (6) in red.

Table 3. P&L characteristics.

Model	μ_1	σ_1	μ_2	σ_2
HEST	0.065	2.298	−0.0644	0.0057
VG-CIR	6.722	20.568	−0.6671	1.647
NIG-CIR	6.986	27.137	−0.732	3.150

moment swap in the hedge of a variance swap leads to a significant improvement of the hedge performance. Not only is the P&L mean much closer to zero, also the standard deviation is much smaller. The tails of the original hedge are much fatter than that of the enhanced hedge. We note that the standard deviation in the Heston case is already quite low using no 3rd moment swaps. This is due to the continuous paths of the model. In the absence of jumps the Taylor-like approximation cut off after the quadratic term in (2) is already very accurate. However real markets jump and it is precisely these jumps that can cause significant hedging errors.

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